sounds in a large auditorium will contain echoes at about 100 millisecond intervals, while 10 milliseconds is typical for a small office. Some species have solved this ranging problem by using *active sonar*. For example, bats and dolphins produce clicks and squeaks that reflect from nearby objects. By measuring the interval between transmission and echo, these animals can locate objects with about 1 cm resolution. Experiments have shown that some humans, particularly the blind, can also use active echo localization to a small extent.

Timbre

The perception of a continuous sound, such as a note from a musical instrument, is often divided into three parts: **loudness**, **pitch**, and **timbre** (pronounced "timber"). *Loudness* is a measure of sound wave intensity, as previously described. *Pitch* is the frequency of the fundamental component in the sound, that is, the frequency with which the waveform repeats itself. While there are subtle effects in both these perceptions, they are a straightforward match with easily characterized physical quantities.

Timbre is more complicated, being determined by the *harmonic content* of the signal. Figure 22-2 illustrates two waveforms, each formed by adding a 1 kHz sine wave with an amplitude of *one*, to a 3 kHz sine wave with an amplitude of *one-half*. The difference between the two waveforms is that the one shown in (b) has the higher frequency *inverted* before the addition. Put another way, the third harmonic (3 kHz) is phase shifted by 180 degrees compared to the first harmonic (1 kHz). In spite of the very different time domain waveforms, these two signals sound *identical*. This is because hearing is based on the *amplitude* of the frequencies, and is very insensitive to their *phase*. The *shape* of the time domain waveform is only indirectly related to hearing, and usually not considered in audio systems.

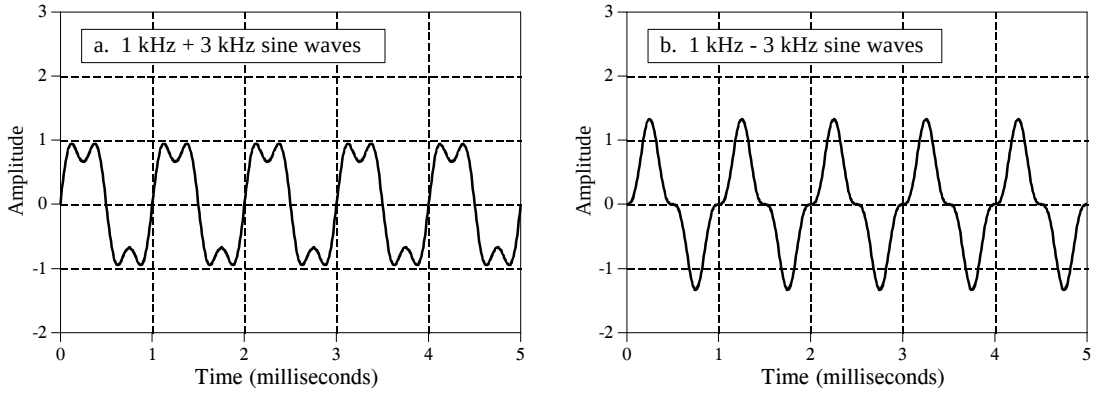

FIGURE 22-2
Phase detection of the human ear. The human ear is very insensitive to the relative phase of the component sinusoids. For example, these two waveforms would sound identical, because the *amplitudes* of their components are the same, even though their relative *phases* are different.